1. What is Azure Portal?

The **Azure Portal** is a **web-based graphical interface** used to manage all Azure services.

It allows administrators to:

- Create and manage resources
- Monitor services
- Configure networking
- Manage security and identity
- Control billing and subscriptions
- Deploy applications

Ways to manage Azure Services:

1. Azure Portal (GUI)
2. Azure CLI
3. PowerShell
4. Cloud Shell

1. What is Azure Resource Manager (ARM)?

Azure Resource Manager (ARM) is the **deployment and management service of Azure**.

It provides a **management layer** that allows you to:

- Create resources
- Update resources
- Delete resources
- Manage dependencies
- Apply policies
- Control access using RBAC

Example Resources Managed by ARM

- Virtual Machines
- Storage Accounts
- Virtual Networks
- Load Balancers
- App Services
- Databases

2. What is an ARM Template?

An **ARM Template** is a **JSON file** that defines the infrastructure and configuration of Azure resources.

Key Idea

Instead of:

> Manually creating VM → Network → Storage → Security

You write a **template**, and Azure deploys everything automatically.

3. Benefits of ARM Templates

1. Infrastructure as Code (IaC)

Infrastructure is written in code.

Example:

```
Plain text

VM configuration
Network configuration
Storage configuration
```

All stored in code.

2. Repeatable Deployments

Deploy same environment many times.

Example:

```
Dev environment
Testing environment
Production environment
```

3. Automation

ARM templates can be used in:

- CI/CD pipelines
- DevOps automation
- Infrastructure provisioning

4. Version Control

Templates can be stored in:

- GitHub
- GitLab
- Azure DevOps

How to Deploy ARM Template

Steps:

1. Go to Azure Portal
2. Search

```
Deploy a custom template
```

3. Upload template file
4. Enter parameters
5. Deploy

1. What are Azure Resource Locks?

Azure Resource Locks are used to **protect important resources**.

They ensure that resources **cannot be deleted or modified accidentally**.

Example resources to protect:

- Virtual Machines
- Resource Groups
- Storage Accounts
- Virtual Networks
- Databases

2. Types of Azure Locks

Azure provides **two types of locks**.

Lock Type	Description
ReadOnly	Resource cannot be modified or deleted
Delete (CanNotDelete)	Resource cannot be deleted but can be modified

1. ReadOnly Lock

This lock makes the resource **read-only**.

Users can:

✔ View resource
✖ Modify resource
✖ Delete resource

Example blocked actions:

- Stopping VM
- Changing VM size
- Updating storage configuration

Example:

```
ReadOnly Lock on Storage Account
```

Even an **Owner cannot modify it**.

2. Delete Lock (CanNotDelete)

This lock **prevents deletion only**.

Users can:

✔ Modify resource
✔ Update configuration
✖ Delete resource

Example:

```
Virtual Machine
Delete Lock applied
```

Admin can:

- Restart VM
- Resize VM

But **cannot delete it**.

3. Lock Scope (Where Locks Apply)

Locks can be applied at **three levels**.

Level	Scope
Subscription	All resources inside subscription
Resource Group	All resources inside RG
Resource	Only specific resource

Example

```
Subscription
    │
    └── Resource Group (Lock applied)
            │
            ├── Virtual Machine
            ├── Storage Account
            └── Virtual Network
```

All resources inside that **resource group become locked**.

4. Lock Inheritance

Locks follow **inheritance rules**.

If lock applied at higher level:

```
Subscription Lock
        ↓
Resource Group
        ↓
Resource
```

All child resources inherit the lock.

Example:

```
Resource Group → ReadOnly Lock
```

Result:

- VM cannot be modified
- Storage cannot be modified
- Network cannot be modified